Implementation of an 8-bit unsigned pipelined multiplier with configurable accuracy

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***Abstract*— This paper presents the schematic-level design of an 8-bit unsigned pipelined multiplier with configurable accuracy using Cadence Virtuoso. The architecture allows switching between accurate and approximate modes to trade off power, delay, and output quality. Pipelining improves throughput by reducing the critical path in the multiplier stages. Simulation results show that approximate modes provide lower power, making the design suitable for low-power and error-tolerant applications. The simulation was carried out on Cadence Virtuoso Layout Suite with 45nm process design kit (PDK).**

I. INTRODUCTION

Multipliers are essential building blocks in digital systems, especially in signal processing, communication, and low-power embedded applications. As device complexity increases, there is a growing demand for arithmetic units that offer high performance while maintaining low power and area overhead. Traditional accurate multipliers provide precise results but often suffer from large critical paths and higher power consumption.

Approximate computing has emerged as an effective technique for error-tolerant applications such as IoT sensor nodes, multimedia processing, and machine learning accelerators. By relaxing accuracy requirements, significant improvements in energy efficiency and speed can be achieved. Combining this concept with pipelining further enhances throughput by breaking the computation into smaller stages, thereby reducing delay and improving operating frequency.

This approach demonstrates the advantages of incorporating configurable accuracy into pipelined arithmetic units for low-power and adaptive digital systems.

Constant	Value
L_{min}	45 nm
W_{min}	120 nm
VDD_{NOM}	1 V

II. ARCHITECTURE

The proposed 8-bit unsigned pipelined multiplier is designed to achieve high throughput and configurable accuracy while maintaining low power consumption. The overall architecture consists of three major computational blocks—Booth Encoder & Selector, Partial Product Addition Tree, and a 16-bit Brent–Kung Adder—organized into pipeline stages for improved performance.

A. Input Register Stage

The 8-bit operands **A[7:0]** and **B[7:0]** are first passed through flip-flops (FFs) to form the initial pipeline boundary.

These registers help stabilize the input signals and ensure synchronized operation for the downstream logic.

B. Booth Encoder and Selector

This block performs radix-4 Booth encoding to reduce the number of generated partial products.

By grouping multiplier bits into overlapping pairs, the Booth unit identifies the required multiples of the multiplicand (0, ±A, ±2A).

This reduces hardware complexity and improves speed compared to a simple array multiplier.

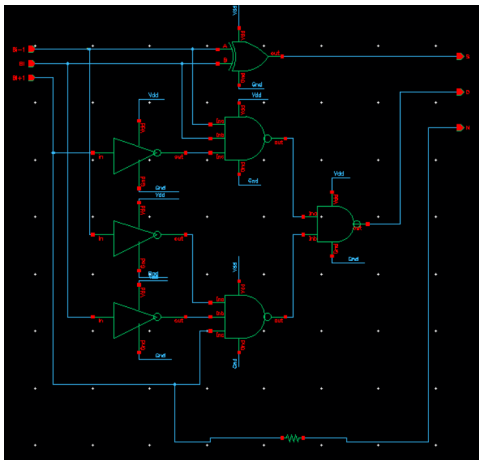


Fig. 1. Booth Encoder

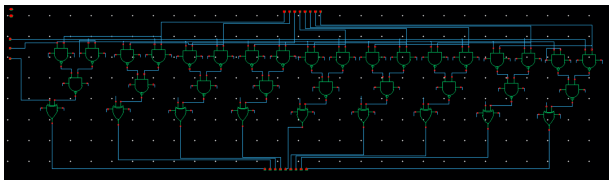


Fig. 2. Booth Selector

C. Partial Product Addition Using Tree Structure

The encoded partial products are summed using a tree-based reduction structure (e.g., Wallace/ Dadda/ compressor

tree). The tree structure significantly reduces the addition delay by compressing multiple partial products in parallel. This block forms the core of the multiplication process and sits inside the first pipeline stage.

D. Pipeline Register

A flip-flop stage is added after the partial product reduction tree. This breaks the long combinational path and creates a clear boundary for **Pipeline Stage-1**, enabling higher clock frequency.

E. 16-bit Brent–Kung Adder

The second major computation block is a 16-bit Brent–Kung adder, chosen for its balanced trade-off between speed, area, and fan-out.

It finalizes the multiplication by performing fast carry-lookahead addition on the reduced partial product outputs.

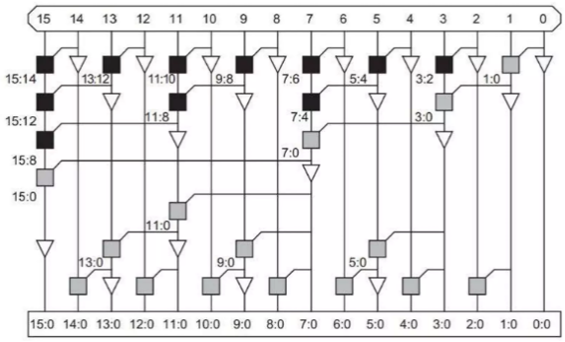


Fig. 3. Brent Kung Adder

This adder forms **Pipeline Stage-2**.

F. Output Register Stage

A final set of flip-flops captures the 16-bit multiplication result **Mul[15:0]**.

This ensures stable output timing and helps maintain high throughput in pipelined operation.

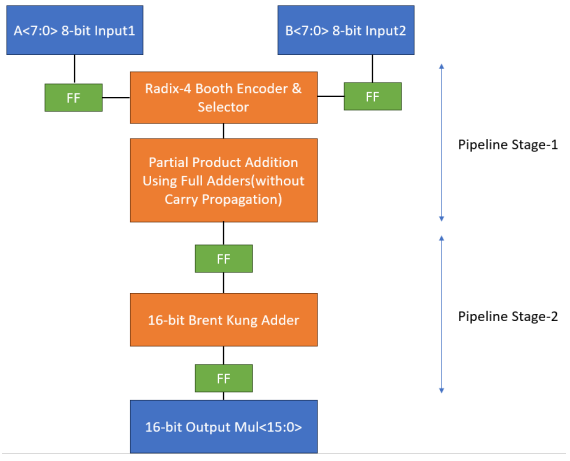


Fig. 4. Architecture of 8 Bit unsigned multiplier

III. Proposed Design for Accuracy Configuration

To improve energy efficiency, the proposed multiplier incorporates **data gating** and **power gating** techniques at the schematic level. These methods help reduce unnecessary switching activity and leakage power, which are critical in low-power digital systems such as IoT and battery-powered devices.

A. Data Gating

Data gating reduces dynamic power by blocking signal transitions when the input data does not need to be processed. In this design, the gating logic monitors the enable/accuracy-mode signals and selectively propagates data only to the active blocks.

*a) **How Data Gating Is Used in This Multiplier**:*

- When the multiplier operates in an **approximate mode**, the last four LSBs of the inputs(A<3:0> , B<3:0>) are disabled.
- Flip-flops at pipeline boundaries receive data only when valid input is available, avoiding unnecessary toggling.
- The Brent–Kung adder receives gated inputs during idle cycles to minimize switching.

*b) **Benefits**:*

- Lower dynamic power consumption.
- Reduced transition activity in combinational blocks.
- No impact on performance since gating is controlled at the clock boundary.

B. Power Gating

Power gating is used to reduce leakage power by disconnecting the supply (VDD or ground) from inactive circuits through a high-Vt header or footer transistor.

*a) **How Power Gating Is Applied in This Design**:*

- Certain blocks involved in generating or compressing lower bits of partial products are power-gated in approximate modes.
- Sleep transistors (header/footer) isolate unused circuitry during low-accuracy operation.
- The gated blocks stay completely off, eliminating leakage.

*b) **Benefits**:*

- Significant reduction in leakage power when approximate or low-activity modes are used.
- Helps achieve substantial power savings in standby or low-precision computation.
- Highly effective for IoT and low-duty-cycle systems.

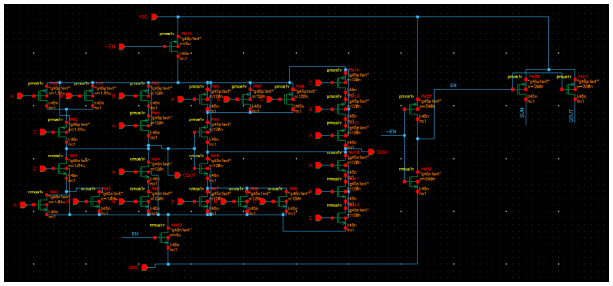


Fig. 5. Power Gated Mirror Full Adder

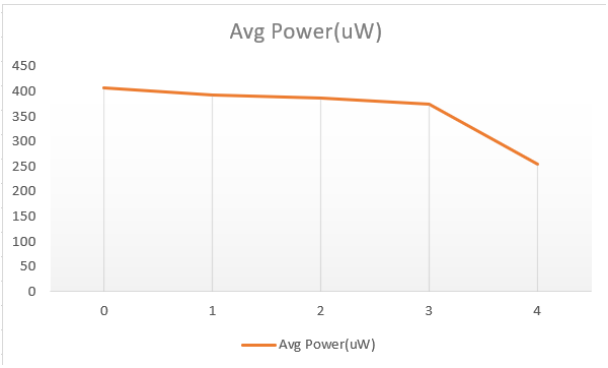


Fig. 6. Avg Power vs Accuracy Mode

IV. PERFORMANCE RESULTS AND CHARECTARIZATION

The power results demonstrate the effectiveness of the configurable-accuracy architecture. In **Mode 0**, all blocks of the multiplier operate at full precision, giving the highest average power of **407.269 µW** and peak power of **2.50 mW**. As accuracy is reduced from Mode 1 to Mode 4, unused partial-product logic and sections of the compressor tree are disabled through **data gating** and **power gating**, which lowers switching activity and leakage.

This results in a gradual decrease in power for Modes 1–3, while **Mode 4** achieves the most improvement with a **37.73% reduction in average power** and **40% reduction in peak power**. The reduction in peak power indicates fewer instantaneous switching events inside the tree structure and adder, improving overall energy efficiency.

Since the design is pipelined, these savings are achieved without affecting throughput, making the multiplier suitable for low-power and error-tolerant applications such as IoT and embedded DSP systems.

Operating Frequency : 500MHz Max Delay : 311.3ps (Mode 0)

TABLE I
POWER VS ACCURACY MODES

Mode	Avg Power (μW)	Peak Power (mW)
0	407.269	2.5
1	392	2.4
2	385.681	2.3
3	374.096	2.26
4	253.598	1.5

V. ERROR ANALYSIS

To evaluate the accuracy degradation introduced by the proposed multi-mode approximate multiplier, three error metrics—**Mean Error**, **RMS Error**, and **Normalized Mean Error Distance (NMED)**—were computed for Modes 1 through 4. Mode 0, representing the fully accurate multiplier, naturally shows **zero error across all metrics** and serves as the reference baseline.

Mode 0 shows zero error and acts as the accurate reference. For the approximate modes (1–4), the error gradually increases as more approximation is applied. **Mode 1** introduces very small error, with low Mean Error, RMS Error, and NMED values, indicating output quality very close to the exact multiplier. **Modes 2 and 3** show moderate increases in error but still maintain low NMED (10 range), making them suitable for most error-tolerant applications.

Mode 4 shows the highest error, as it uses the most aggressive approximation to achieve maximum power savings. Although the Mean Error and NMED are significantly larger, this mode is still useful for low-power applications where exact precision is not critical.

Overall, the results clearly show the expected **accuracy–power trade-off**, where higher approximation reduces power but increases computational error.

TABLE II
ERROR METRICS FOR DIFFERENT ACCURACY MODES(ABSOLUTE VALUES)

Mode	Mean Error	RMS Error	NMED
1	2.888888889	4.4721	4.44E-05
2	16.7059	27.5	2.73E-04
3	16.65	27.5	2.53E-04
4	8259.85	9100	0.112

VI. CONCLUSION

A reconfigurable multiplier enabling scalable accuracy–power trade-offs has been presented. The design employs data gating and power gating across five operational modes. Mode 0 provides exact computation at 407 µW. Modes 1–3 maintain high accuracy (NMED ĩ0–10) while reducing power, with Mode 3 achieving 374 µW. Mode 4 maximizes energy savings (253 µW, 38% reduction) at elevated

error (NMED 0.112), targeting error-tolerant applications. The architecture demonstrates effective dynamic adaptation between computational precision and energy consumption within a unified hardware platform.

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